

Technical Analysis of IPv4 Fragmentation and Reassembly

Summary

IPv4 fragmentation is a Network Layer mechanism that enables a datagram to traverse links with smaller Maximum Transmission Units (MTUs).

When:

Datagram Size > MTU

and DF = 0 → the router fragments the packet.

Fragmentation depends on three IPv4 header fields:

- Identification
- Flags (DF, MF)
- Fragment Offset

A critical architectural rule is the **8-byte scaling constraint** of the Fragment Offset field. Because Fragment Offset is 13 bits while Total Length is 16 bits, offsets are expressed in units of 8 bytes. Therefore:

All fragments except the last must carry data divisible by 8 bytes.

Reassembly occurs only at the destination host.

1. Fragmentation in Layered Context

Different layers perform different splitting operations:

Layer	Data Unit	Process
Application	Message	—
Transport	Segment	Segmentation
Network	Datagram	Fragmentation

Data Link Frame Framing

Important distinction:

At the Network Layer, the entire Transport segment (including its header) is treated as raw data.

2. IPv4 Fragmentation Mechanics

Fragmentation is triggered when a router forwards a packet to a link with a smaller MTU.

2.1 Key Header Fields

1. Identification (16 bits)

- Unique per original datagram.
- All fragments share the same ID.
- Used for grouping fragments during reassembly.

2.2 Flags (3 bits)

- Bit 0: Reserved
- Bit 1: DF (Don't Fragment)
- Bit 2: MF (More Fragments)

DF = 1

- Router must not fragment.
- If packet exceeds MTU → dropped.

MF = 1

- More fragments follow.

MF = 0

- This is the last fragment (or only fragment).

2.3 Fragment Offset (13 bits)

Indicates:

Number of data bytes preceding this fragment

But stored in units of 8 bytes:

Offset field value = (Bytes before this fragment) / 8

3. The 8-Byte Scaling Rule

Why scaling is required:

- Total Length → 16 bits → max 65,535 bytes
- Fragment Offset → 13 bits → max 8,191

To cover full range:

$8,191 \times 8 \approx 65,528$

Thus:

Offset is measured in multiples of 8 bytes.

Mandatory Constraint

Every fragment except the last must have:

Data size % 8 = 0

If not divisible:

Router rounds down to nearest multiple of 8.

4. Analytical Example

Given:

Original Data = 500 bytes

Header = 20 bytes

Total Length = 520 bytes

MTU = 200 bytes

Step 1: Maximum Data per Fragment

$MTU - \text{Header} = 200 - 20 = 180 \text{ bytes}$

Step 2: Apply 8-byte Rule

$180 / 8 = 22.5 \rightarrow \text{Not integer}$

Nearest lower multiple:

176 bytes

Step 3: Fragment Creation

Fragment 1

- Data = 176
- Header = 20
- Offset = 0
- MF = 1

Fragment 2

- Data = 176
- Header = 20
- Offset = $176 / 8 = 22$

- MF = 1

Fragment 3 (Last)

Remaining data:

$500 - 176 - 176 = 148$ bytes

- Data = 148
- Header = 20
- Offset = $352 / 8 = 44$
- MF = 0

Note:

Last fragment does not need data divisible by 8.

5. Reassembly at Destination

Only the destination host performs reassembly.

Routers never reassemble.

Why Routers Do Not Reassemble

1. Fragments may take different paths.
2. Reassembly consumes memory and CPU.
3. A fragment may need further fragmentation.
4. Router bottlenecks must be avoided.

5.1 Reassembly Logic

Receiver performs:

Step 1: Identify First Fragment

Offset = 0

Step 2: Identify Last Fragment

MF = 0

Step 3: Verify Continuity

If:

Current Offset = X

Data length = L

Next expected offset:

$X + (L / 8)$

Example:

Offset = 0

Data = 176

Next expected offset:

$0 + 22 = 22$

5.2 Out-of-Order Handling

Fragments may arrive out of order.

Receiver:

- Uses Identification field to group fragments.
- Uses Fragment Offset to place them correctly.
- Uses MF to detect completion.

6. Fragment Loss Consequences

If any one fragment is lost:

- Reassembly fails.
- Receiver discards all fragments of that datagram.
- Source must retransmit entire original datagram.

Reason:

Source does not know fragmentation occurred.
It retransmits at Transport Layer level (e.g., TCP).

7. Performance and Efficiency Analysis

7.1 Header Overhead

If n fragments are created:

Extra header overhead:

$$(n - 1) \times 20 \text{ bytes}$$

Total header transmitted:

$$n \times 20 \text{ bytes}$$

7.2 Efficiency Formula

Efficiency:

Original Data

Data + All Fragment Headers

Example (3 fragments)

Data = 500 bytes

Headers = $3 \times 20 = 60$ bytes

Total transmitted = 560 bytes

Efficiency:

$500 / 560 \approx 89.29\%$

7.3 Throughput Impact

Throughput depends on:

Efficiency $\times$ Bottleneck Bandwidth

Fragmentation reduces efficiency $\rightarrow$ lowers effective throughput.

8. Critical Architectural Insights

1. Fragmentation is Data-Centric

Only data is divided.

Header is copied and modified.

2. IPv4 vs IPv6

IPv4:

- Routers and source can fragment.

IPv6:

- Only source can fragment.
- Routers drop oversized packets.

3. Last Fragment Exception

Only final fragment can violate 8-byte rule because:

No future offset depends on it.

4. Routers Never Merge Fragments

Even if next link has large MTU:

Fragments remain separate.

Only destination reassembles.

